MODULE 2 · LECTURE 6

CS 6530 Applied Cryptography

Block Cipher Modes of Operation

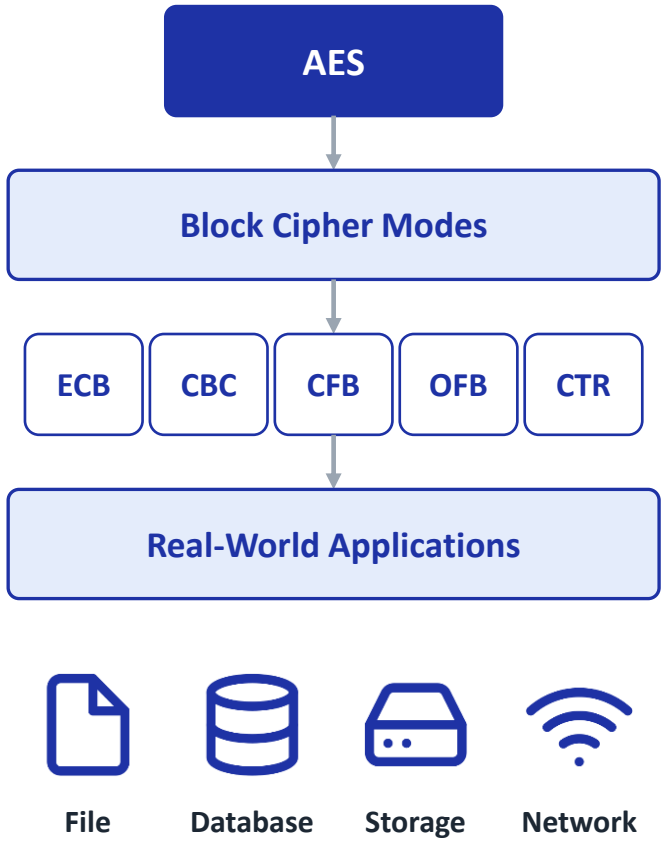
Module 2 – Symmetric Cryptography, Hashes and Authentication

LEARNING OUTCOMES

Block Cipher Modes of Operation

By the end of this lecture, you will be able to:

- Explain why block cipher modes of operation are required.
- Describe the five classical block cipher modes and their purpose.
- Compare ECB, CBC, and CTR based on security and engineering properties.
- Explain the role of IVs, nonces, and padding.
- Select an appropriate block cipher mode for different application scenarios.



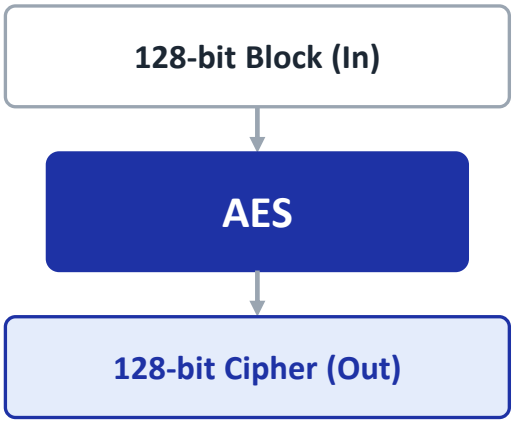
A block cipher encrypts one fixed-size block at a time. Modes of operation define how it securely encrypts larger data such as files, databases, storage volumes, and network traffic.

MOTIVATION

AES Encrypts One Block at a Time

AES Block Cipher

- Input: one 128-bit plaintext block
- Output: one 128-bit ciphertext block
- Fixed block size irrespective of AES-128, AES-192, or AES-256



One Block In → One Block Out

Engineering Insight

AES block size is always 128 bits. Only the key size changes (128, 192, or 256 bits).

Real-World Data = Many Blocks



→ **Thousands (or Millions) of Blocks**

AES encrypts only one fixed-size block at a time. Modes of operation specify how multiple blocks are securely processed to encrypt real-world data.

ENGINEERING DESIGN

Evolution of Block Cipher Modes

ENGINEERING PROBLEM

"How should multiple plaintext blocks be encrypted securely?"

Different applications have different engineering requirements.

Engineering Design Question	Primary Design Goal	Example Application
Independent encryption	**Simplicity**	Independent block processing
Previous ciphertext dependency	**Hide plaintext patterns**	Sequential file encryption
Parallel encryption	**High throughput & random access**	Storage systems, databases
Stream-cipher behaviour	**Continuous data encryption**	Network traffic, streaming

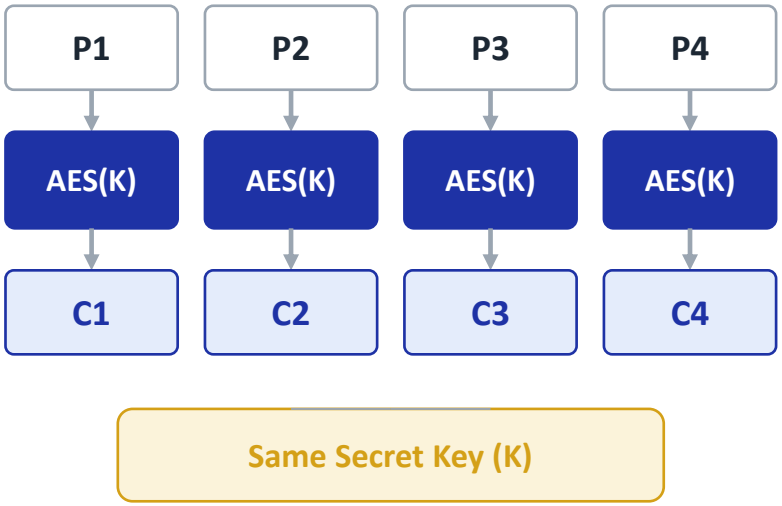
These engineering requirements led to different block cipher modes of operation

ECB | **CBC** | **CFB** | **OFB** | **CTR**

Different engineering requirements led to different block cipher modes. Each mode offers distinct security, performance, and usability characteristics for practical applications.

MODE OF OPERATION

Electronic Codebook (ECB)



Key Characteristics

- Each plaintext block is encrypted independently.
- The same secret key is used for every block.
- No IV or nonce is required.
- Encryption and decryption can be parallelized.

Engineering Observation

Independent block processing makes ECB simple to implement and naturally supports parallel processing.

Compact Operational Model

Encryption: $C_i = E_k(P_i)$

Decryption: $P_i = D_k(C_i)$

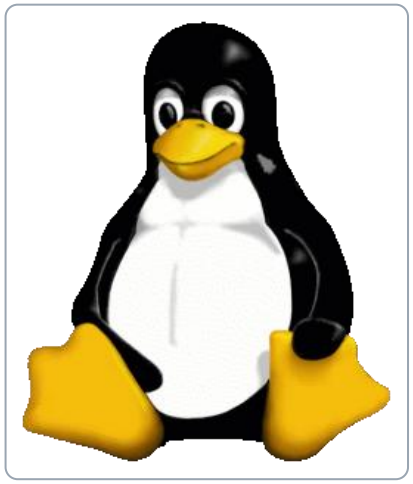
Think Like an Engineer

If two plaintext blocks are identical, what relationship would you expect between the corresponding ciphertext blocks?

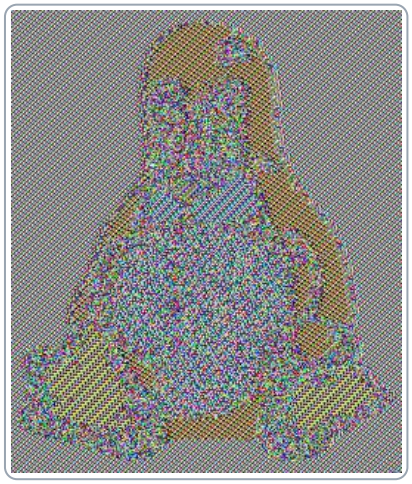
ECB encrypts each plaintext block independently using the same secret key. Its simplicity enables straightforward implementation and parallel processing.

OBSERVE

ECB Pattern Leakage – The Penguin Demonstration



Original (Plaintext)



Encrypted – AES, ECB Mode

Compact Operational Model

Identical plaintext blocks + same key → identical ciphertext blocks.

Repeated regions preserve visible structure.

Think Like an Engineer

If AES is secure, why can we still recognize the object?

Engineering Insight: *A strong cryptographic algorithm must also be used in a mode appropriate to the application.*

ECB preserves equality between identical plaintext blocks; repeated structures can therefore remain visible after encryption.

Image source: “Tux” the Linux penguin, originally created by Larry Ewing. Used for educational demonstration of AES-ECB pattern leakage.

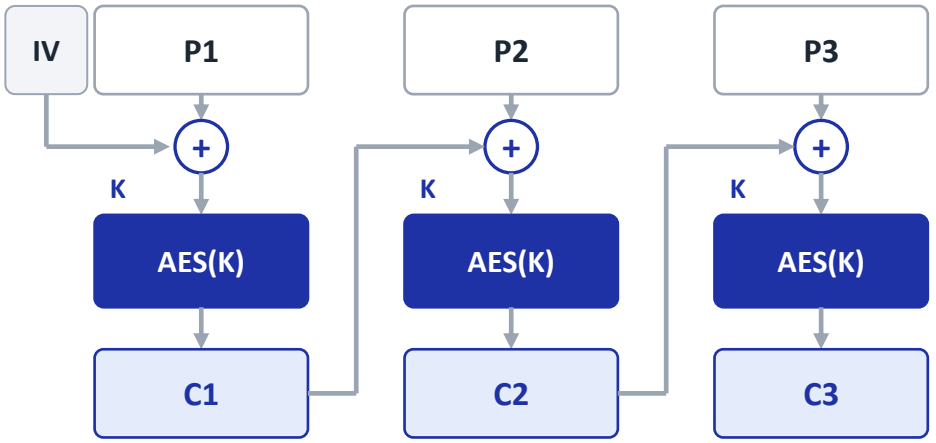
ENGINEERING SOLUTION

Cipher Block Chaining (CBC)

Engineering Principle

A deterministic block cipher must be combined with changing context so that each encryption operation receives a unique

AES is deterministic



Chained: each ciphertext block feeds the next XOR

Key Teaching Points

- Each ciphertext block depends on all previous plaintext blocks.
- Identical plaintext blocks no longer produce identical ciphertext.
- An IV supplies the initial context (C_0 = IV).

Compact Operational Model

Encryption: $C_i = E_k(P_i \oplus C_{i-1})$, $C_0 = IV$

Decryption: $P_i = D_k(C_i) \oplus C_{i-1}$

Think Like an Engineer

If one ciphertext block is corrupted during transmission, which plaintext blocks will be affected? Explain using the decryption equation.

CBC introduces changing context by combining each plaintext block with the previous ciphertext block before encryption, ensuring identical plaintext blocks are presented differently to AES.

ENGINEERING EVALUATION

Engineering Considerations in CBC

Engineering Reflection

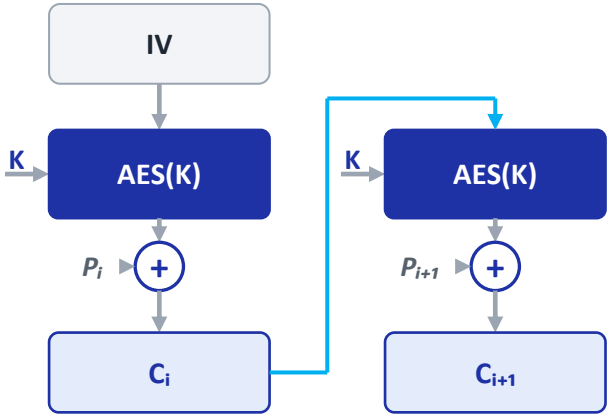
Engineering solutions are evaluated not only by the problems they solve, but also by the operational considerations they introduce.

Engineering Aspect	Evaluation
Pattern Leakage	Removed – identical plaintext blocks no longer produce identical ciphertext.
Initialization Vector (IV)	Required; must be unpredictable and sent alongside the ciphertext.
Padding	Required when plaintext length is not a multiple of the block size.
Encryption	Inherently sequential – each block depends on the previous ciphertext block.
Decryption	Can be parallelized – all ciphertext blocks are already available.
Error Propagation	A single bit error affects two plaintext blocks during decryption.

CBC successfully removes ECB's repeated-pattern problem while introducing practical engineering considerations that designers must understand before deployment.

Cipher Feedback (CFB) and Output Feedback (OFB)

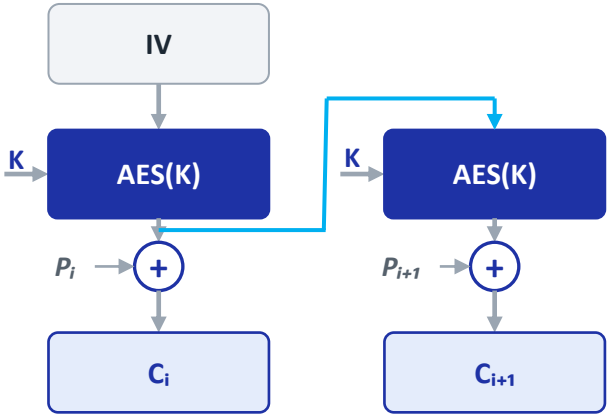
Cipher Feedback (CFB)



Feedback: previous CIPHERTEXT re-encrypted

CFB feeds ciphertext back into AES to generate the next keystream block.

Output Feedback (OFB)



Feedback: previous AES OUTPUT re-encrypted

OFB feeds AES's own output back into AES, independent of the ciphertext.

Compact Operational Model

CFB: $C_i = P_i \oplus E_k(C_{i-1})$, $C_o = IV$

OFB: $O_i = E_k(O_{i-1})$; $C_i = P_i \oplus O_i$, $O_o = IV$

Think Like an Engineer

Both modes generate a keystream. What is the key architectural difference in how the next keystream value is produced?

CFB and OFB are classical stream-like block cipher modes that differ primarily in the information fed back to generate the next keystream. Modern systems more commonly employ CTR and authenticated encryption modes such as GCM.

MODERN STREAM-LIKE MODE

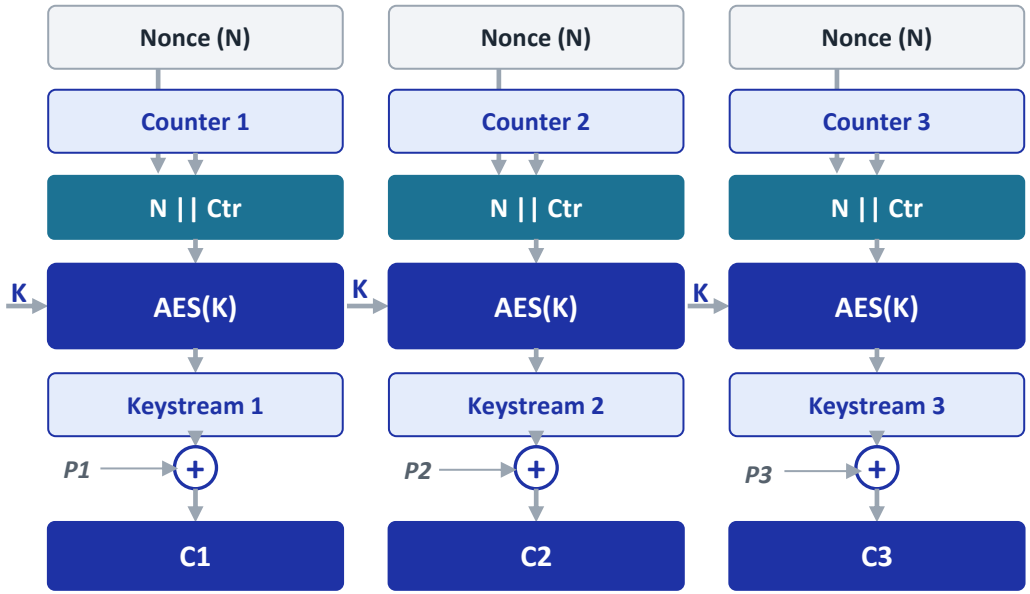
Counter (CTR) Mode

Engineering Motivation

Instead of obtaining changing context from previous ciphertext, CTR constructs a unique input for every AES invocation using a nonce and a counter.

CBC vs CTR: Source of

CBC	Previous Ciphertext
CTR	**Nonce + Counter** No mandated split size



Three independent, parallel AES(K) operations

NIST Special Publication 800-38A — Recommendation for Block Cipher Modes of Operation: Methods and Techniques

Compact Operational Model

Keystream: $KS_i = E_k(N \,||\, i)$

Encryption: $C_i = P_i \oplus KS_i$

Decryption: $P_i = C_i \oplus KS_i$

Think Like an Engineer

Why does CTR not require the AES decryption operation during decryption?

CTR generates a unique keystream by encrypting nonce–counter values. The keystream is XORed with the plaintext, enabling independent processing of each block and forming the foundation of AES-GCM.

ENGINEERING EVALUATION

Engineering Properties of Counter (CTR) Mode

Engineering Reflection

CTR removes inter-block dependencies, allowing encryption and decryption of each block independently – highly suitable for modern computing platforms.

Security Warning

The same nonce–counter sequence must never be reused under the same key.

Engineering Property	Engineering Implication	Typical Applications
Parallel Processing	Encryption and decryption of each block are fully independent.	*High-throughput networking, hardware acceleration*
No Padding	Ciphertext length exactly matches plaintext length.	*Streaming data, variable-length messages*
Random Access	Any block can be decrypted directly without processing prior blocks.	*Encrypted storage, random-access files*
Limited Error Propagation	A bit error affects only the corresponding plaintext block.	*Noisy or lossy transmission channels*
Nonce Uniqueness	The nonce–counter pair must never repeat under the same key.	*Security-critical protocols (e.g., TLS, AES-GCM)*

Compact Operational Reasoning

$P_i = C_i \oplus E_k(N \,||\, i)$

Bit error in C_i → only C_i affected → only P_i affected → others unaffected

Think Like an Engineer

Both CBC and CTR can decrypt a selected block without decrypting the entire message. What additional information does CBC require compared with CTR?

CTR combines high performance, parallelism, random access, and minimal error propagation, making it one of the most widely used block cipher modes. Its security depends critically on never reusing the same nonce–counter sequence under the same key.

ENGINEERING BUILDING BLOCKS

IVs, Nonces, and Padding

Engineering Perspective

Although ECB, CBC, and CTR all use AES, each mode requires different supporting mechanisms to operate securely and efficiently.

Concept	Used By	Engineering Purpose
# IV	CBC, CFB, OFB	Ensures each encryption run starts from a fresh, unpredictable state.
Nonce	CTR	Combined with a counter to guarantee a unique AES input per block.
Padding	ECB, CBC	Extends plaintext to a full block boundary before encryption.

CTR does not require padding – the generated keystream can be XORed with plaintext of any length.

Compact Operational Connection

CBC

$C_0 = IV$

IV initializes the chaining process.

CTR

$KS_i = E_k(N \,||\, i)$

Nonce provides unique counter inputs.

Padding

13-byte P → 16-byte block

Padding fills to the block boundary.

Think Like an Engineer

Why can CTR encrypt plaintext of arbitrary length without requiring padding, even though AES still processes fixed-size blocks?

IVs, nonces, and padding are essential supporting mechanisms that enable different block cipher modes to achieve secure and efficient operation in real-world systems.

Engineering Comparison of Block Cipher Modes

Engineering Perspective

Different block cipher modes represent different engineering trade-offs, chosen based on security, performance, and operational requirements.

Legend

- Engineering Advantage
- Partial Support / Caveat
- Engineering Limitation

Engineering Property	ECB	CBC	CFB	OFB	CTR
Requires Initialization	●	● *IV*	● *IV*	● *IV*	● *Nonce*
Padding-Free Operation	●	●	●	●	●
Supports Parallel Encryption	●	●	●	●	●
Supports Parallel Decryption	●	●	●	●	●
Resists Pattern Leakage	●	●	●	●	●
Supports Random Access	●	●	●	●	●
Typical Applications	Legacy / Education	File Encryption	Byte-Stream Protocols	Stream-like Data	**High-Perf, Parallel Systems**

CTR stands out by combining multiple engineering advantages for many modern systems.

Think Like an Engineer

You are designing a high-speed encrypted cloud storage system requiring random access, parallel processing, and protection against pattern leakage. Which mode would you choose? Why?

Selecting a block cipher mode is an engineering design decision based on security requirements, performance, and operational constraints – not simply the encryption algorithm itself.

Engineering Applications of Block Cipher Modes

Engineering Perspective

The choice of block cipher mode is driven by the operational profile of the application – not just its need for confidentiality.

Engineering Insight

XTS-AES is a specialized NIST-approved mode for storage-device encryption, optimized for fixed-size sectors and preferred over general network modes for full-disk encryption.

Application	Typical Mode	Why This Mode
Legacy File Encryption	ECB	Simple, independent block processing (education / legacy only).
Streaming Data	CFB / OFB	Converts AES into a stream-like cipher for continuous data.
Database Fields / Records	CTR	Enables random access to individual encrypted fields.
Network Traffic	CTR	Supports high-throughput, parallel encryption of packets.
TLS / IPsec / QUIC	AES-GCM	Authenticated encryption – confidentiality + integrity.
Storage Volumes / Disk	XTS-AES	Specialized mode for fixed-size storage sector encryption.

AES-GCM and XTS-AES are introduced here as application examples, not fully explained.

Think Like an Engineer

You are designing full-disk encryption for a laptop. Would CTR be your first choice? Why or why not?

Different applications require different block cipher modes. Modern network protocols commonly use AES-GCM, while storage systems employ specialized modes such as XTS-AES.

ENGINEERING OUTLOOK

Looking Beyond Classical Block Cipher Modes

AES-GCM

Next Lecture →

Authenticated Encryption with Associated Data (AEAD)

- Confidentiality
- Integrity
- Authentication

XTS-AES

A specialized, NIST-approved mode built for storage encryption – protecting fixed-size disk sectors with efficient random access.

Other Block Ciphers

AES finalists and lightweight ciphers for constrained devices:

- Twofish
- Serpent
- RC6
- PRESENT
- GIFT

Engineering Insight: *Cryptographic engineering extends beyond choosing an algorithm – engineers must also select an appropriate mode and, in some cases, specialized cipher designs.*

Think Like an Engineer

Why might an IoT sensor, a cloud server, and an encrypted laptop use different cryptographic constructions, even though all need confidentiality?

AES remains today's dominant block cipher, but modern cryptography continues to evolve through authenticated encryption, specialized storage modes, and lightweight cipher designs.

LECTURE WRAP-UP

Summary and Looking Ahead

Key Takeaways

- A block cipher encrypts one fixed-size block; modes of operation extend it to real data.
- ECB is simple but leaks patterns – identical blocks produce identical ciphertext.
- CBC removes pattern leakage by chaining ciphertext into each new block.
- CFB and OFB convert AES into stream-like ciphers using different feedback paths.
- CTR generates a keystream from a nonce and counter, enabling parallelism and random access.
- The right mode is chosen through engineering trade-offs – not the algorithm alone.

Learning Roadmap

Lecture 5
AES

Lecture 6
Block Cipher Modes (Today)

Lecture 7
Stream Ciphers: Salsa20, ChaCha20

Lecture 8
Authenticated Encryption: AES-GCM, ChaCha20-Poly1305

Engineering Reflection: *A strong encryption algorithm alone does not create a secure system. Security depends equally on selecting an appropriate mode and using it correctly.*

Block cipher modes transform AES from a secure cryptographic primitive into practical secure communication and storage systems. Engineering success comes from choosing the right mode for the right application.